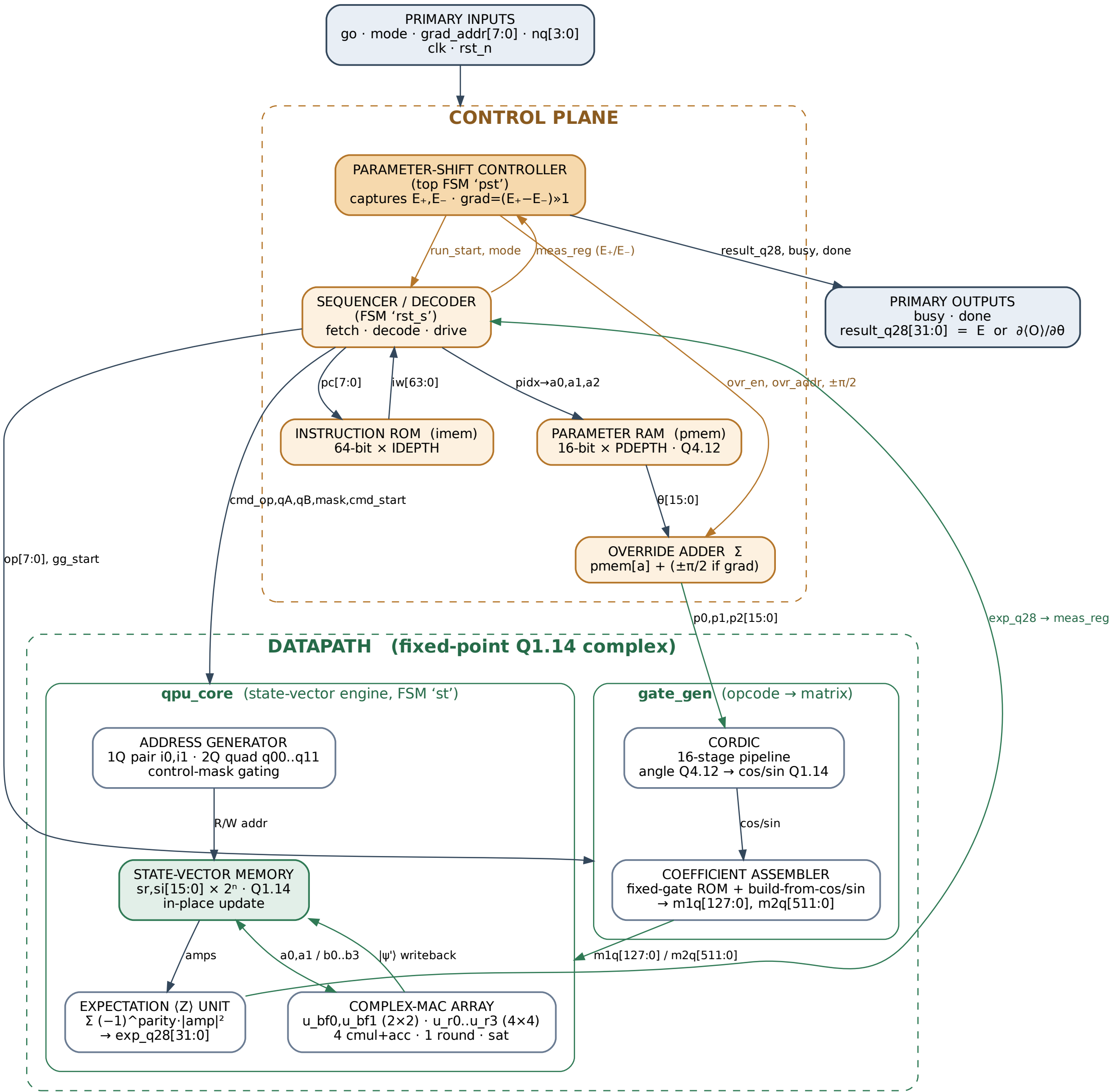


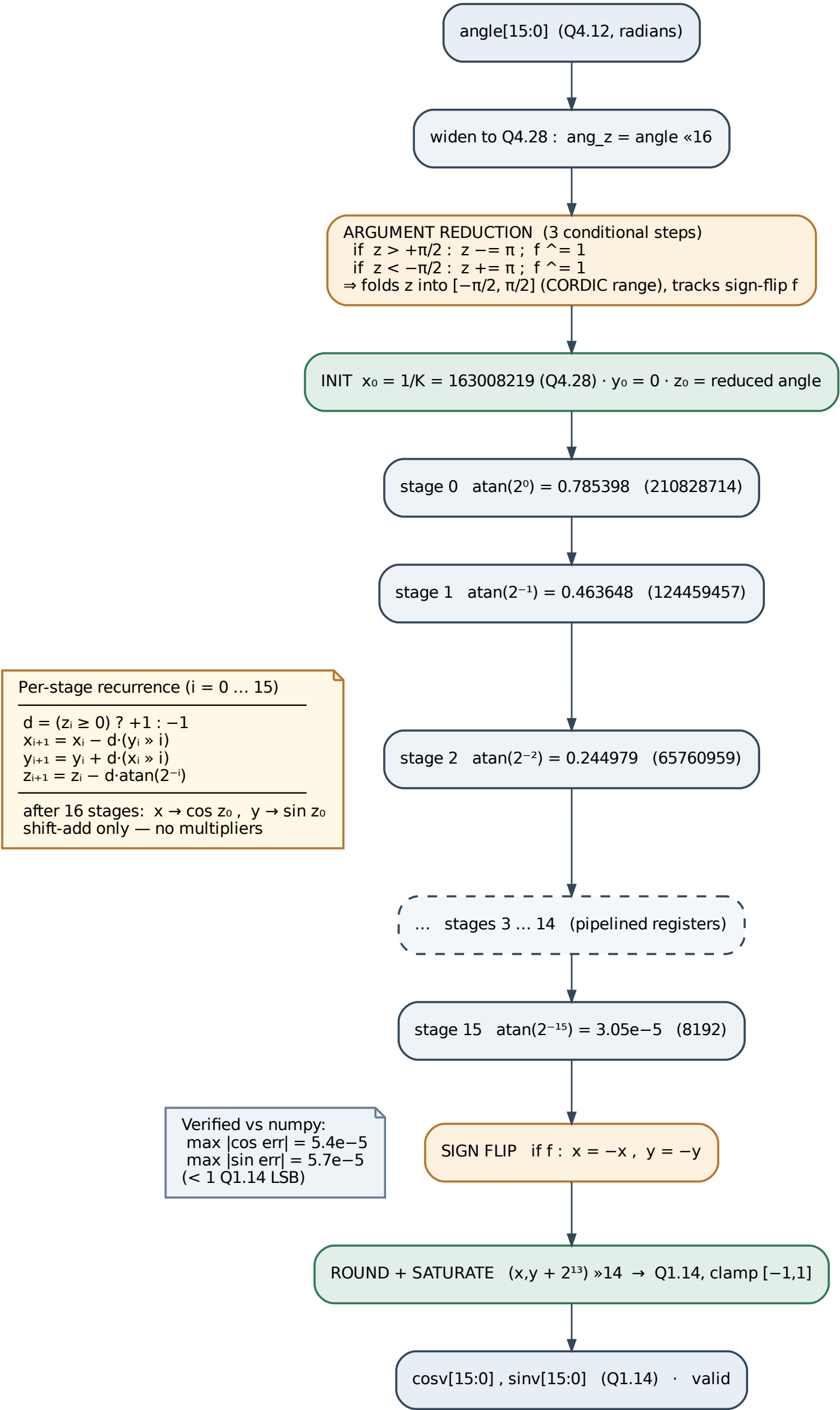
# QML Hardware Accelerator — Top-Level RTL Block Diagram

state-vector engine · module connectivity with signal names and bus widths



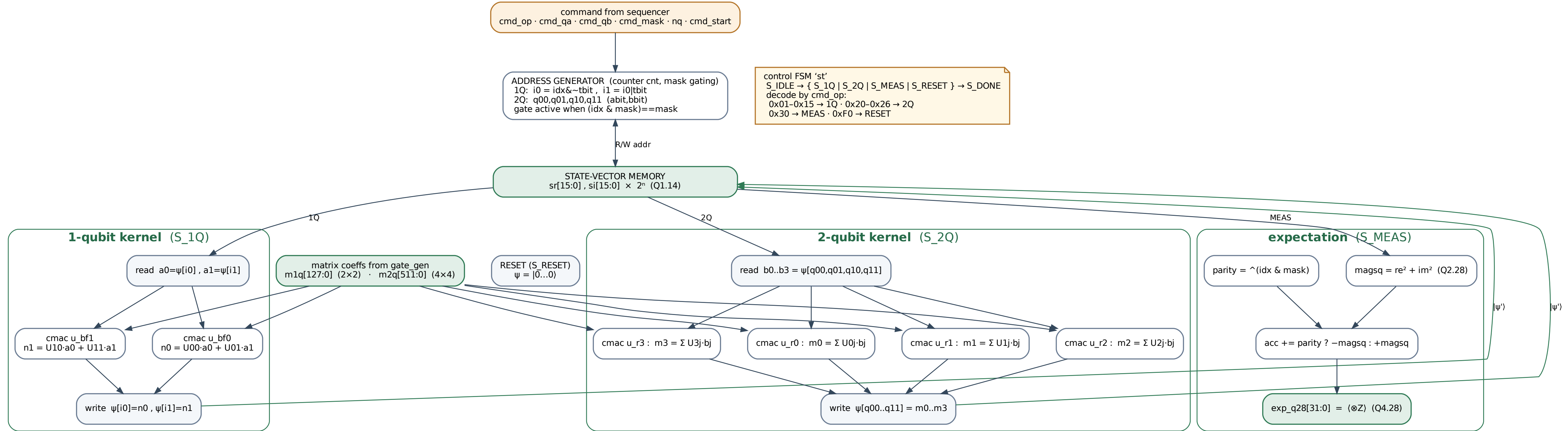
# CORDIC — Pipelined cos/sin Datapath

circular rotation mode · 16 stages · Q4.28 internal · latency = 18 cycles



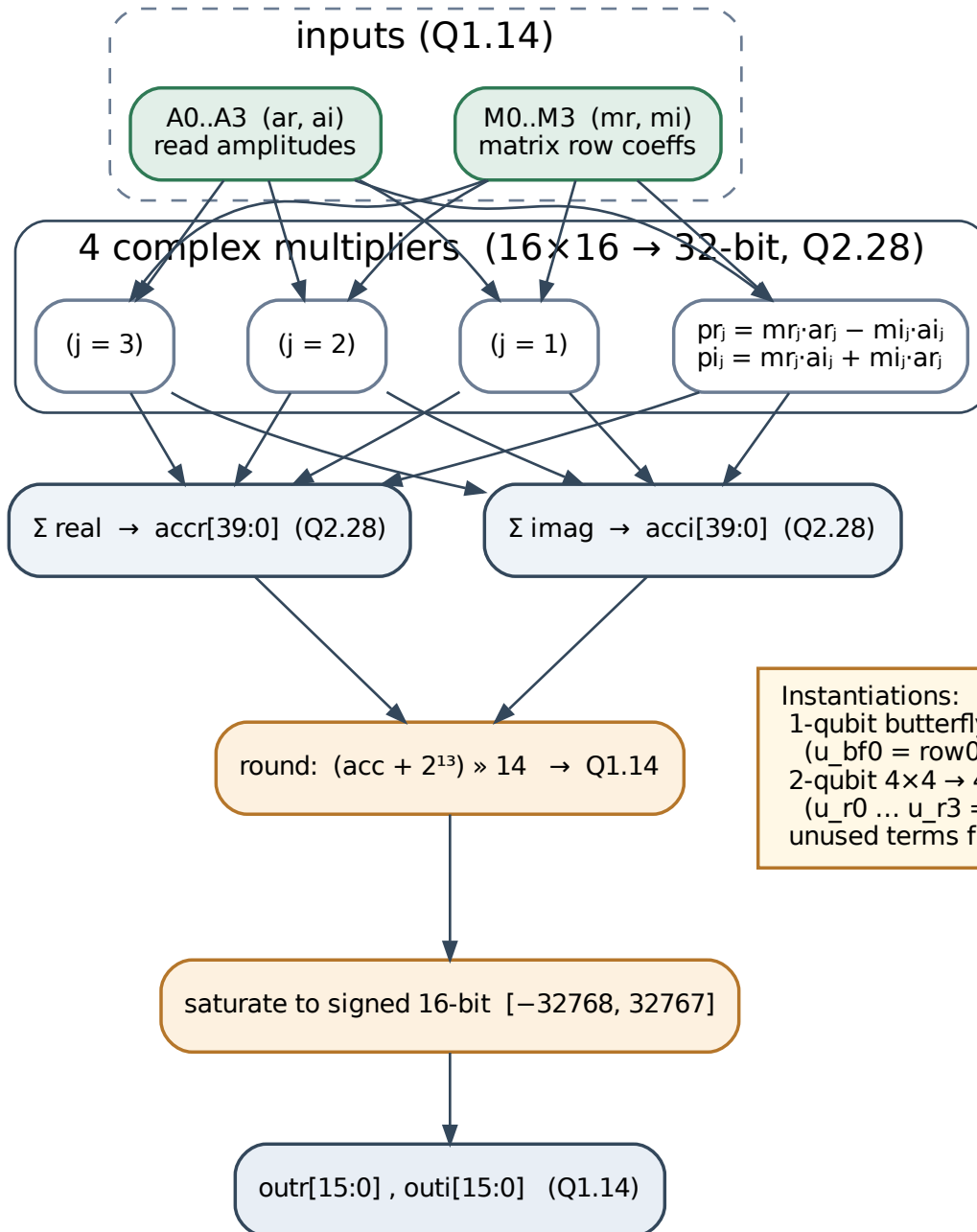
# qpu\_core — State-Vector Datapath

1-qubit butterfly · 2-qubit 4×4 kernel · ⟨Z⟩ expectation · one amplitude(-quad) per clock



# cmac — Complex Multiply-Accumulate Cell

out =  $\sum_j M_j \cdot A_j$  ( $j=0..3$ ) · full-width products · single rounding · saturating



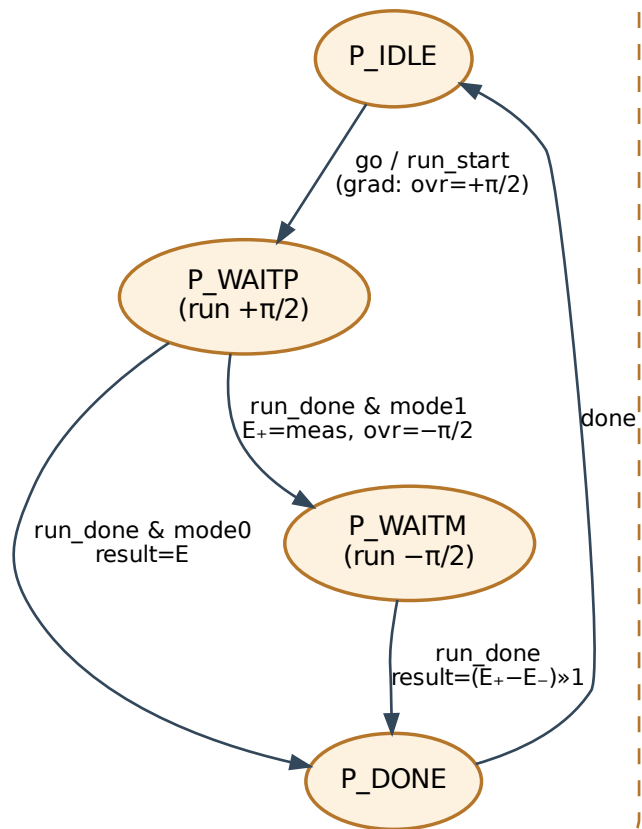
## Instantiations:

1-qubit butterfly → 2 active terms  
(u\_bf0 = row0, u\_bf1 = row1)  
2-qubit 4×4 → 4 active terms  
(u\_r0 ... u\_r3 = rows 0..3)  
unused terms fed M<sub>j</sub> = 0

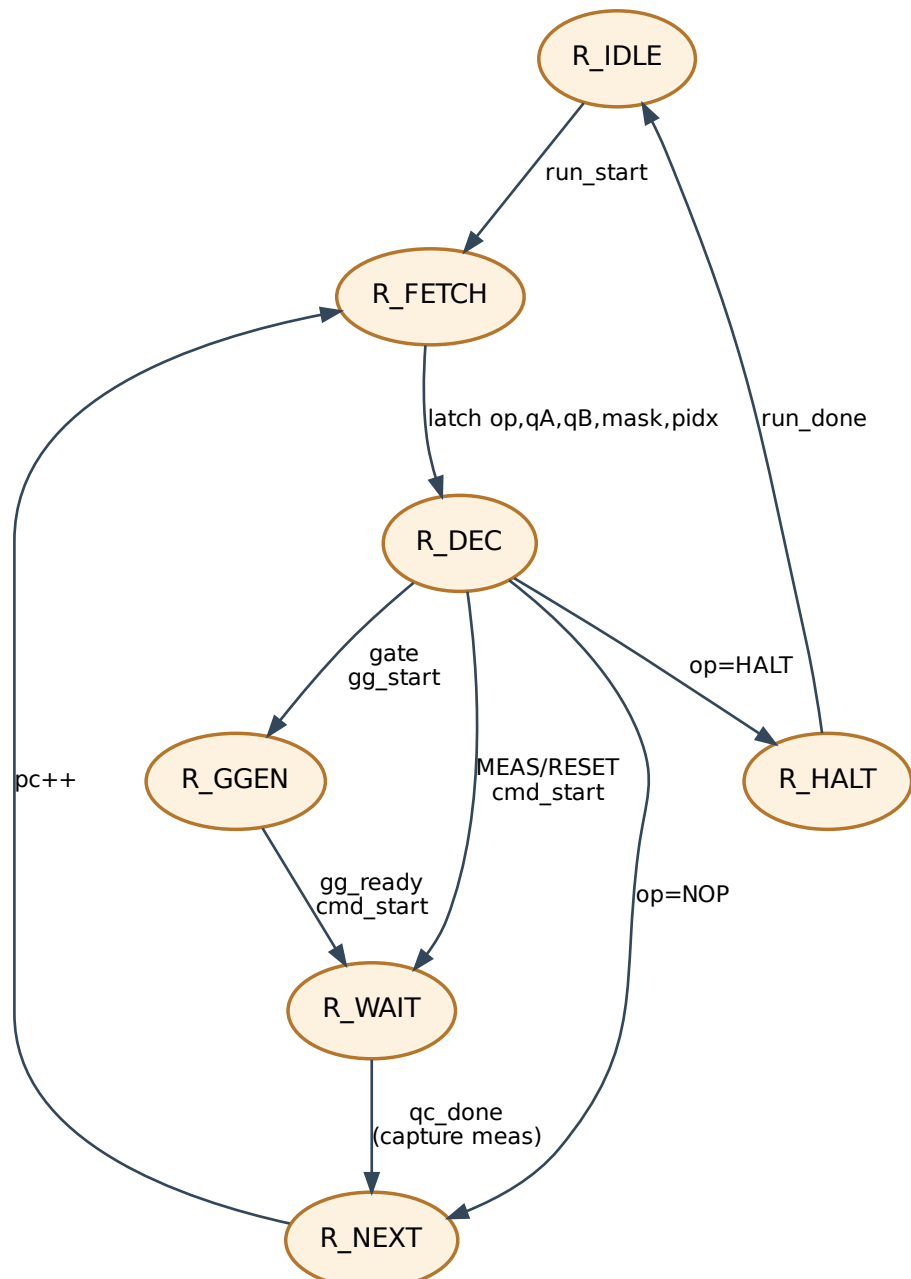
# Control Finite-State Machines

parameter-shift controller · sequencer · gate\_gen · qpu\_core

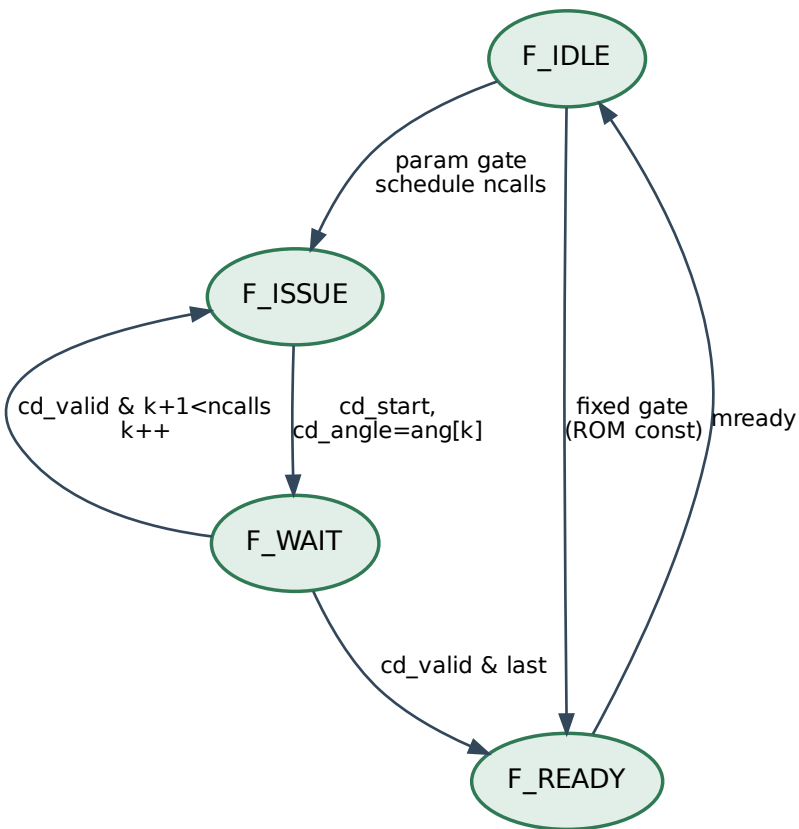
## Parameter-Shift Controller (pst)



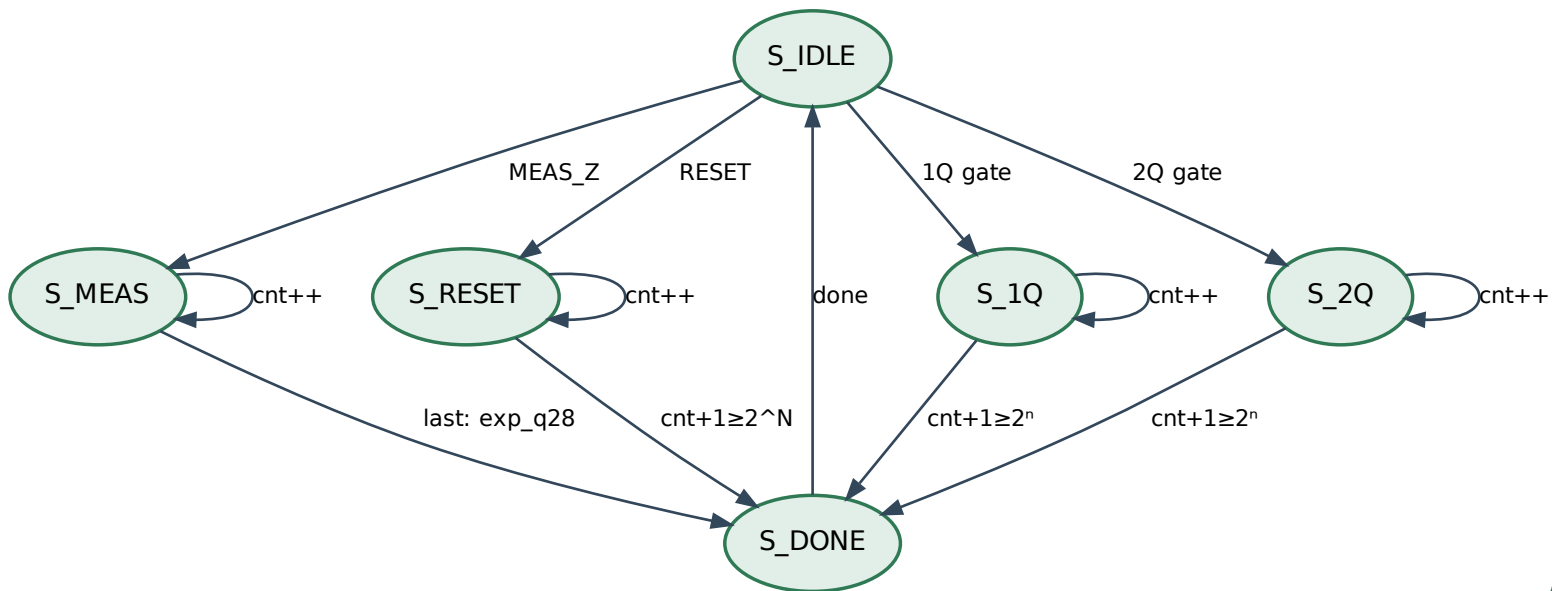
## Sequencer / Decoder (rst\_s)



## gate\_gen (fst)



## qpu\_core (st)



# Hardware Components per Module

what the RTL instantiates — memories, registers, multipliers, muxes, counters, FSMs (5-qubit build)

## CONTROL PLANE

| qml_accelerator (top / sequencer + PSR) |       |                                      |
|-----------------------------------------|-------|--------------------------------------|
| Instruction ROM                         | imem  | 64-bit × 256                         |
| Parameter RAM                           | pmem  | 16-bit × 64                          |
| Program counter                         | pc    | 8-bit counter                        |
| Instruction latch                       | cur_* | op/qa/qb/mask/pidx                   |
| Result registers                        |       | meas_reg, E+, E−, result — 32-bit ×4 |
| Override adders (±π/2)                  |       | 3 adders + 3 compare-muxes           |
| Gradient path                           |       | subtractor + »1 shift                |
| Sequencer FSM                           | rst_s | 8 states                             |
| Param-shift FSM                         | pst   | 6 states                             |

drives

## DATAPATH

| gate_gen (opcode → matrix) |          |                             |
|----------------------------|----------|-----------------------------|
| Opcode latch               | op_r     | 8-bit                       |
| Angle regs                 | ang[0:3] | 4 × 16-bit                  |
| cos/sin store              | cc,ss    | 8 × 16-bit                  |
| Coefficient qmul           |          | 1 multiplier (U2/U3)        |
| Fixed-gate ROM             |          | constants (X,H,S,T,SWAP...) |
| Matrix-select mux          |          | 8 fields (2×2) + 32 (4×4)   |
| Call counter               | k        | 3-bit                       |
| FSM                        | fst      | 4 states                    |

instantiates

| cordic (cos/sin engine)         |          |                       |
|---------------------------------|----------|-----------------------|
| Pipeline x,y,z                  | xr,yr,zr | 17 × 32-bit each      |
| Flags                           | fr,vr    | 17 × 1-bit each       |
| arctan ROM                      |          | 16 × 32-bit constants |
| Per-stage add/sub               |          | 3 × 16 stages         |
| Per-stage shifters              |          | 2 × 16 stages         |
| Sign muxes                      |          | 16 (one per stage)    |
| Arg-reduce + round/sat          |          | 3 add/sub + rounding  |
| no multipliers — shift/add only |          |                       |

| qpu_core (state-vector engine) |         |                          |
|--------------------------------|---------|--------------------------|
| State memory                   | sr, si  | 16-bit × 32, ×2 banks    |
| Sweep counter                  | cnt     | 6-bit                    |
| Accumulator                    | acc     | 48-bit                   |
| Expectation out                | exp_q28 | 32-bit                   |
| MAC cells                      |         | 6 × cmac (see below)     |
| Measurement  amp ²             |         | 2 multipliers + adder    |
| Parity tree                    |         | XOR reduction            |
| Read / write-back muxes        |         | 1Q pair / 2Q quad select |
| FSM                            | st      | 6 states                 |

instantiates ×6

| cmac (×6 instances) |  |                        |
|---------------------|--|------------------------|
| Signed multipliers  |  | 16 × (16×16-bit)       |
| Adder tree          |  | real + imag accumulate |
| Round + saturate    |  | Q2.28 → Q1.14          |
| u_bf0,u_bf1 (2×2)   |  | u_r0...u_r3 (4×4)      |

## TOTAL (5-qubit build)

|                                      |                            |
|--------------------------------------|----------------------------|
| Multipliers (16×16)                  | ≈ <b>98</b> (16×6 MAC + 2) |
| Adders / subtractors                 | ≈ 100                      |
| Flip-flops                           | ≈ 2,300 (CORDIC ≈ 1,700)   |
| Memory                               | ≈ 18.4 kbit                |
| MAC cells · FSMs · counters          | 6 · 4 · 3                  |
| scales with qubits (2 <sup>n</sup> ) | only the memories          |